Loop Elimination in Process Graphs is Confluent when Pruning Steps are Added

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https://clegra.github.io

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Scuola Universitaria Superiore

L'Aquila, Italy

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Universidade de Lisboa

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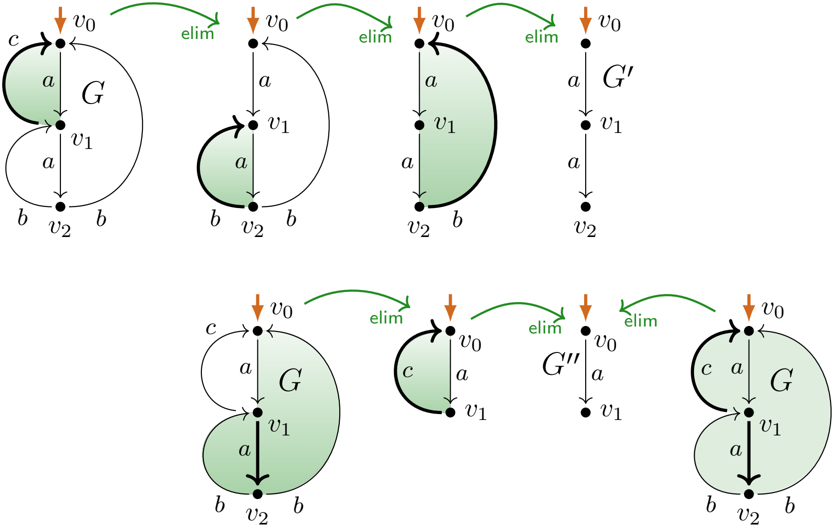
Overview

- Loop elimination $\rightarrow_{\mathrm{elim}}$ in process graphs
  - Milner's process interpretation P of regular expressions
  - Loop existence and elimination property LEE
    LEE $\triangleq$ interpretations of $(*/\cancel{1})$ reg. expr's
  - $\rightarrow_{\mathrm{elim}}$ is not confluent
- Completion of loop elimination $\rightarrow_{\mathrm{elim}}$ with pruning steps $\rightarrow_{\mathrm{prune}}$
  - 'critical pair' lemma for 'bi-loop' elimination steps
  - use of decreasing diagrams
- Application 1: LEE is decidable in polynomial time
- Application 2: Layered LEE = LEE
- Further questions

Loop elimination $\to_{\mathrm{elim}}$



Process interpretation P of regular expressions *(Milner, 1984)*

$\mathbf{0} \quad \overset{P}{\longmapsto}$ deadlock $\boldsymbol{\delta}$, no observable behaviour

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Process semantics $[\![\cdot]\!]_P$ of regular expressions *(Milner, 1984)*

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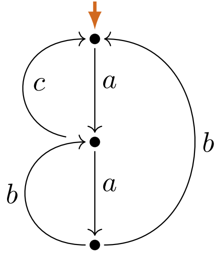
$[\![e]\!]_P := [P(e)]_{\underline{\leftrightarrow}}$ (bisimilarity equivalence class of process $P(e)$)

Process interpretation P (example)

G



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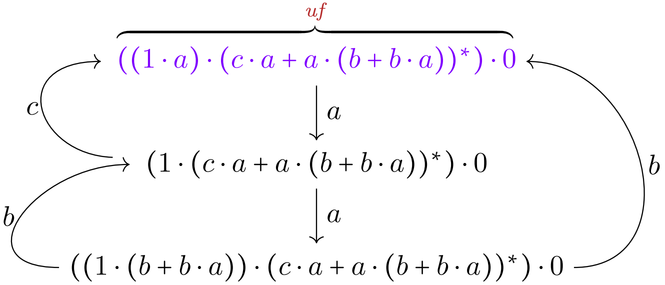

Process interpretation P (example) has property LEE

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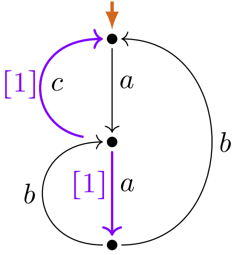

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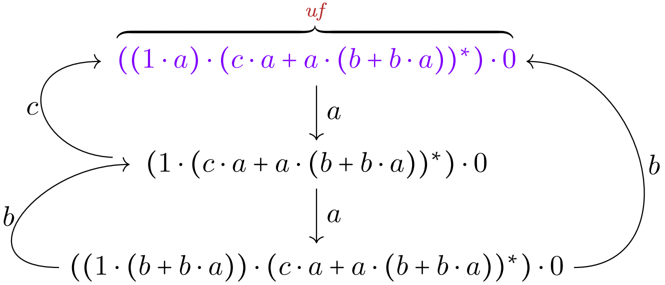

Loop graphs (interpretations of innermost iterations without 1)

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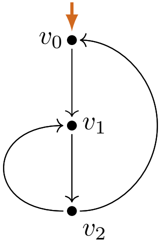

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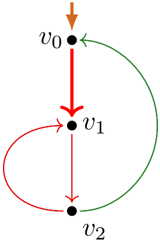

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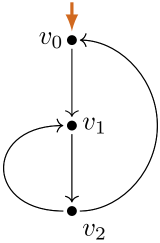

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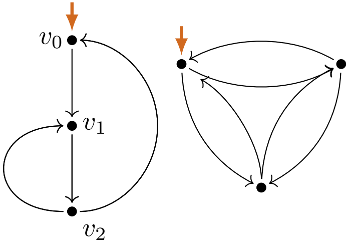

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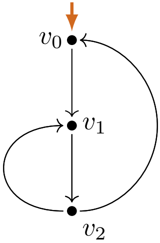

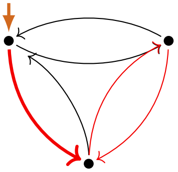

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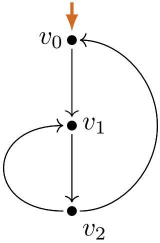

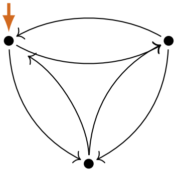

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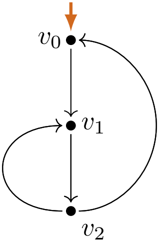

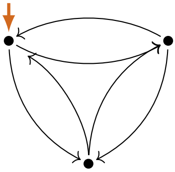

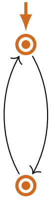

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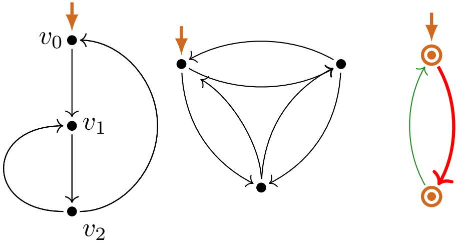

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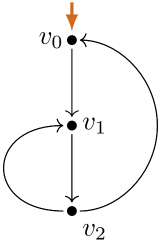

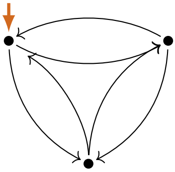

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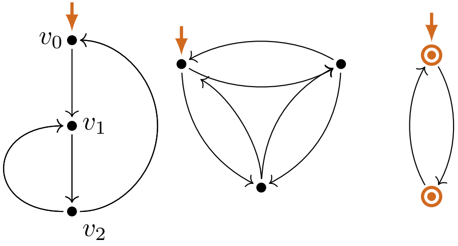

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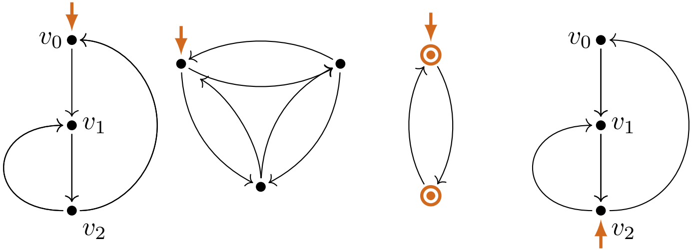

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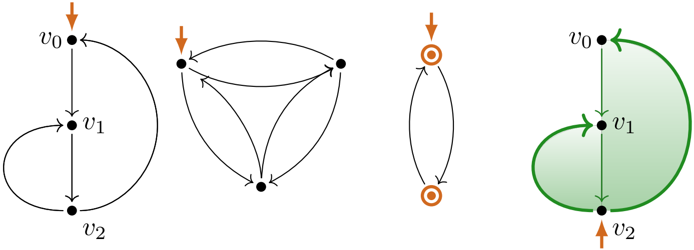

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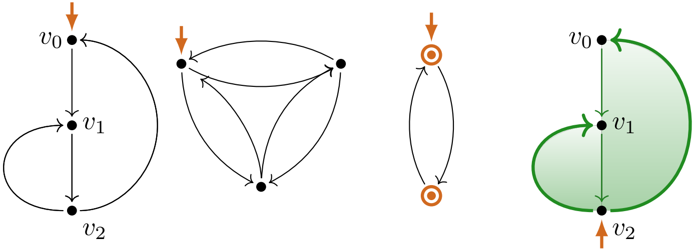

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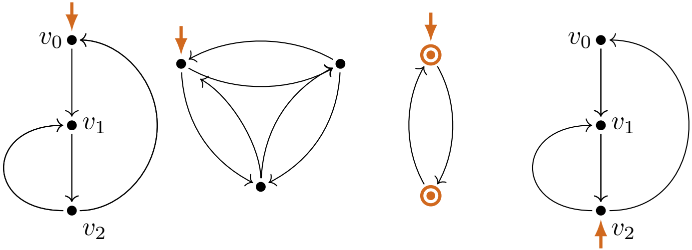

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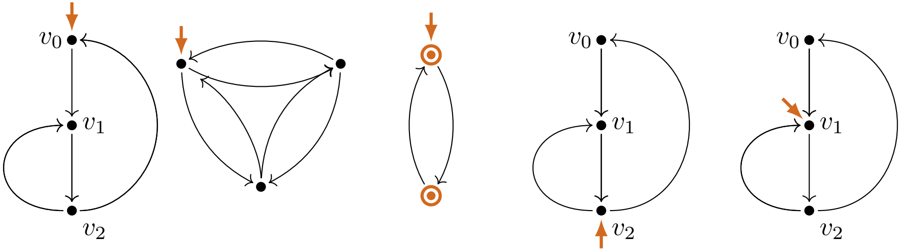

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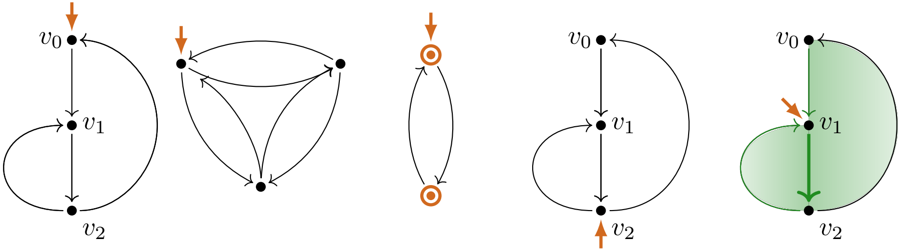

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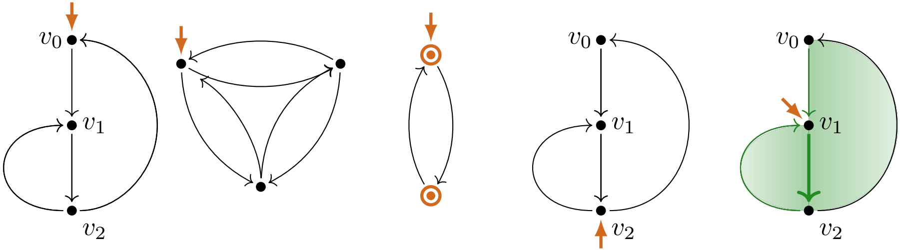

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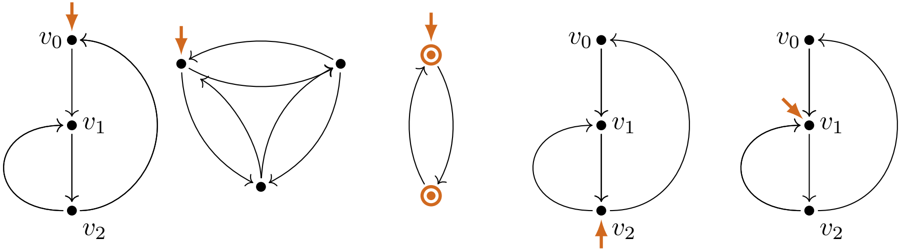

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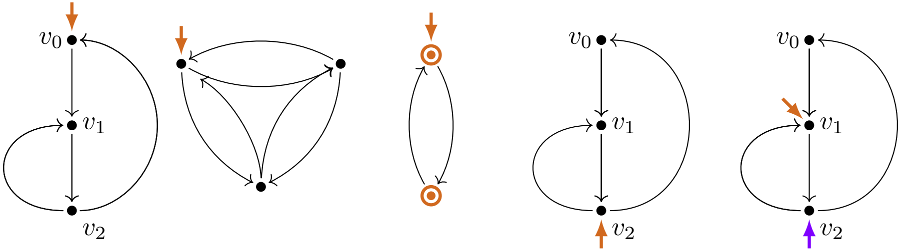

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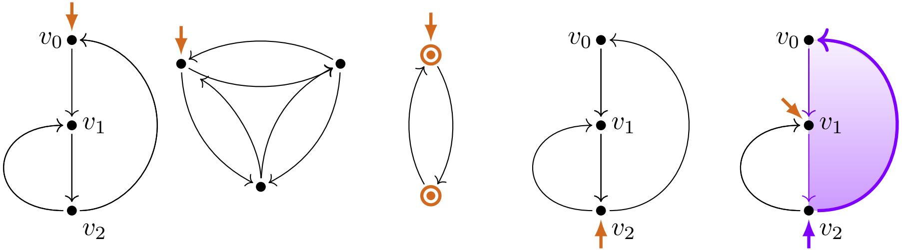

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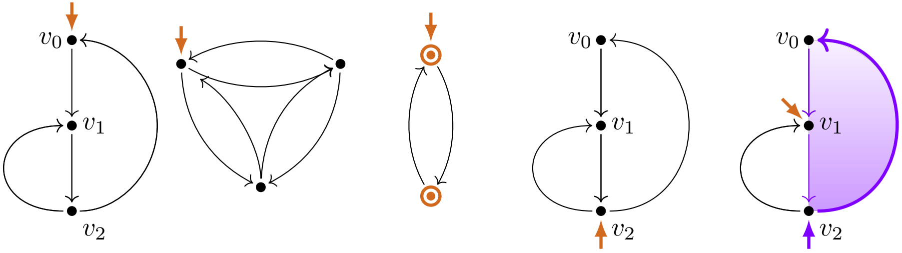

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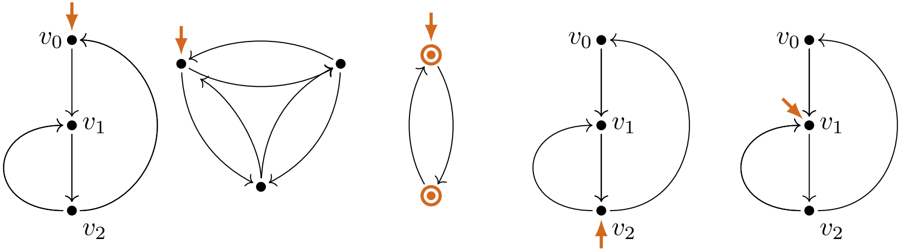

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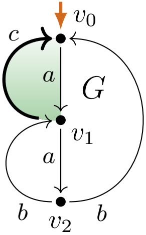

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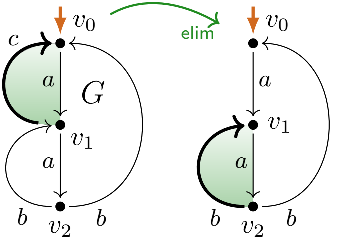

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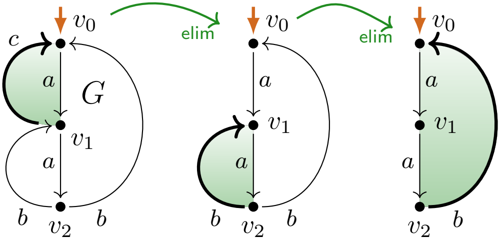

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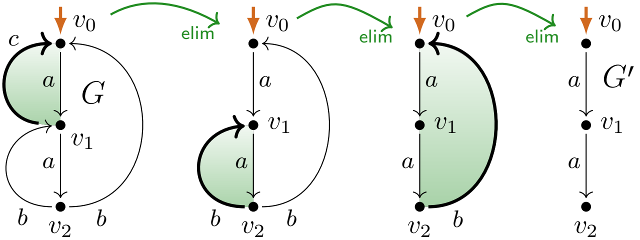

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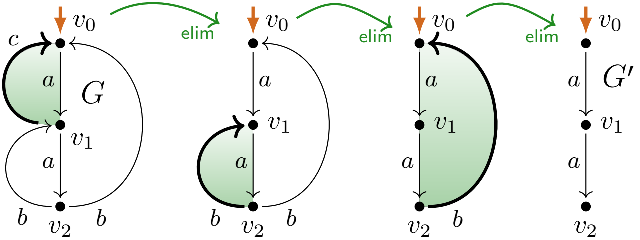

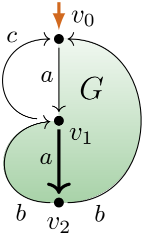

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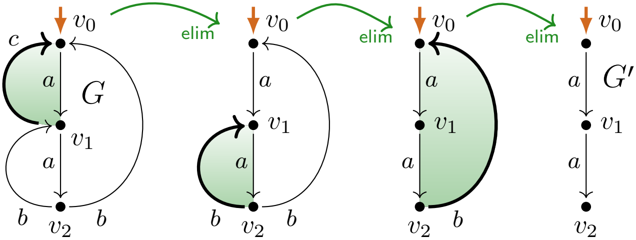

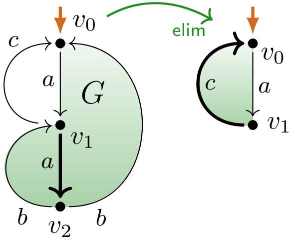

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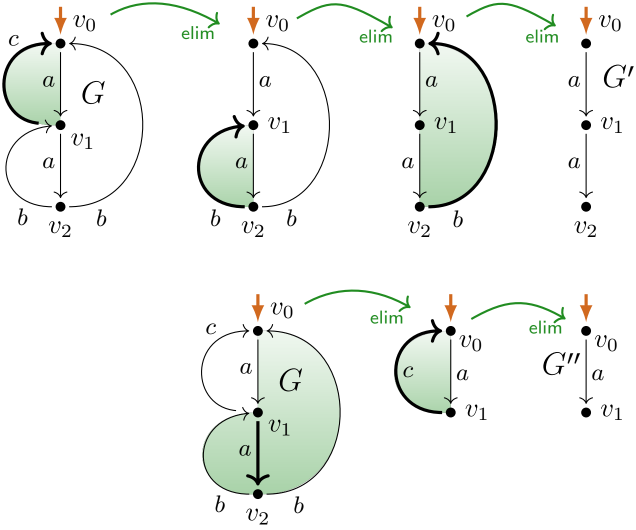

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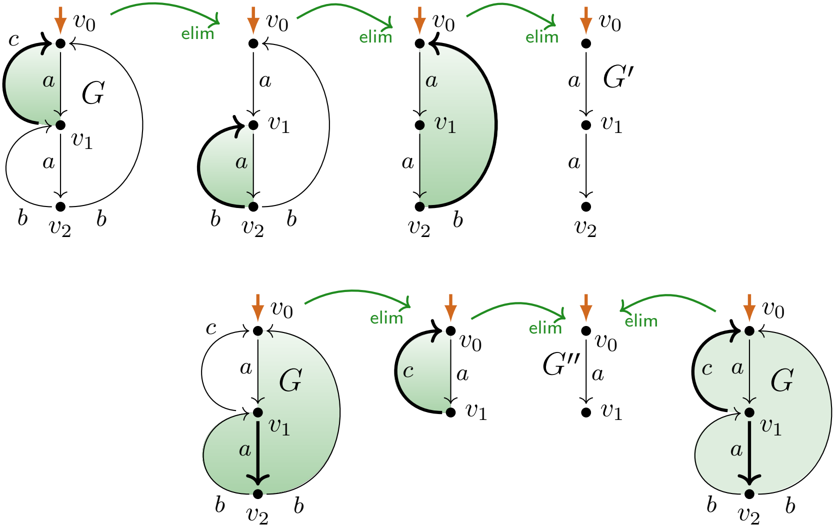

LEE

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$$\exists G_0 \left(G \rightarrow^*_{\text{elim}} G_0 \not\rightarrow_{\text{elim}} \;\wedge\; \underbrace{\neg\infty(G_0)}_{G_0 \text{ permits no infinite trace}} \right).$$

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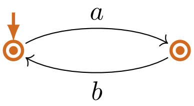

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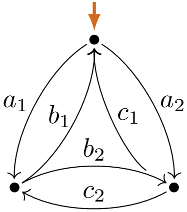

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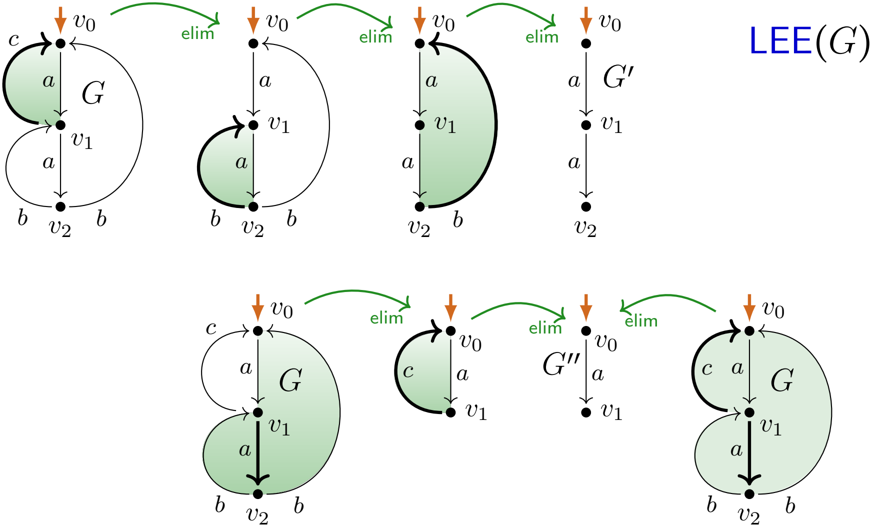

LEE $\hat{=}$ P-/$P^{\bullet}$-expressible graphs for $(*/\!\!\!1)$ reg. expr's

Lemma (*G–Fokkink, LICS'20*)

For all process graphs G:

$$\mathsf{LEE}(G) \implies \exists\, uf \in \mathit{RExp}^{(*/\!\!\!1)}\, \big(\underbrace{G \leftrightarrow P(uf)}_{G \,\in\, [\![uf]\!]_{P}} \big)\,,$$

$$\mathsf{LEE}(G) \impliedby \exists\, uf \in \mathit{RExp}^{(*/\!\!\!1)}\, \big(G \simeq P(uf) \big)\,.$$

LEE $\hat{=}$ P-/$P^{\bullet}$-expressible graphs for $(*/\!\!\!\perp)$ reg. expr's

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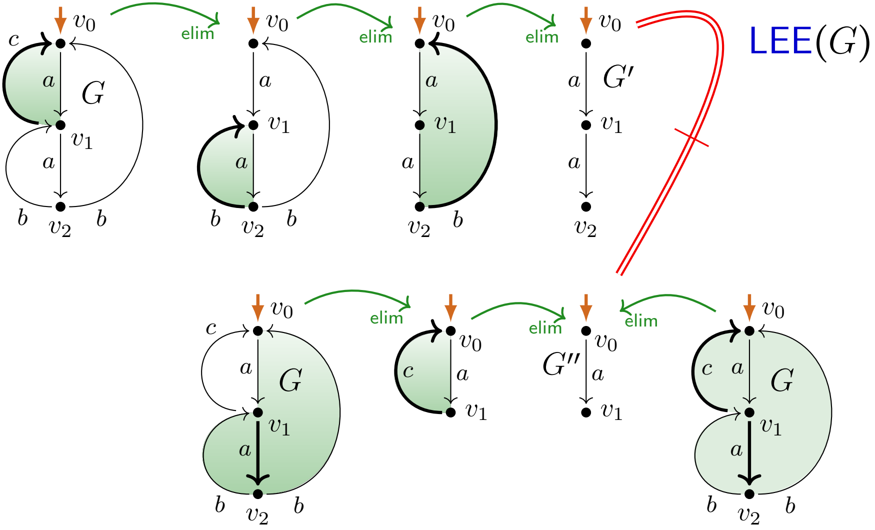
Theorem (*TERMGRAPH '26*, LEE = LLEE *from IWC'26*)

LEE $\hat{=}$ compact process interpretations $P^{\bullet}(uf)$ of $(*/\!\!\!\perp)$ reg. expr's uf:

$$\mathsf{LEE}(G) \iff \exists\, uf \in \mathit{RExp}^{(*/\!\!\perp)} \big(G \simeq P^{\bullet}(uf) \big).$$

Loop elimination $\rightarrow_{\mathrm{elim}}$ is **not** confluent



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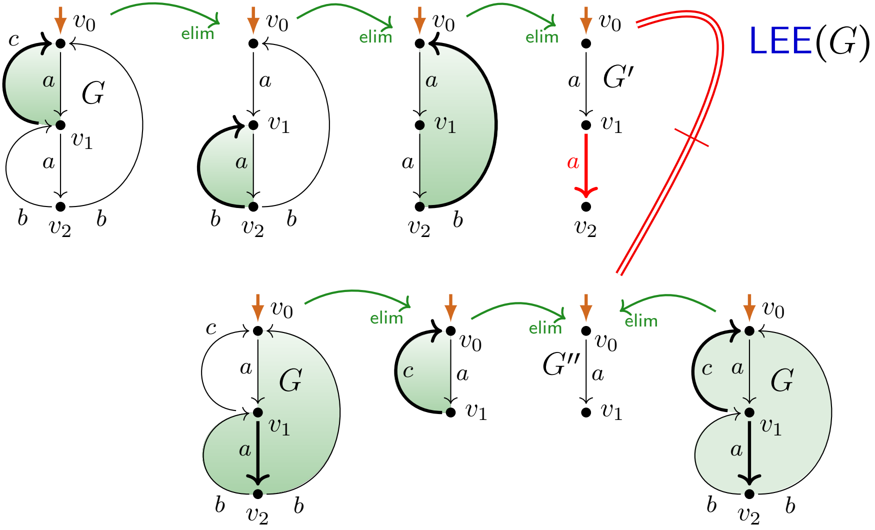
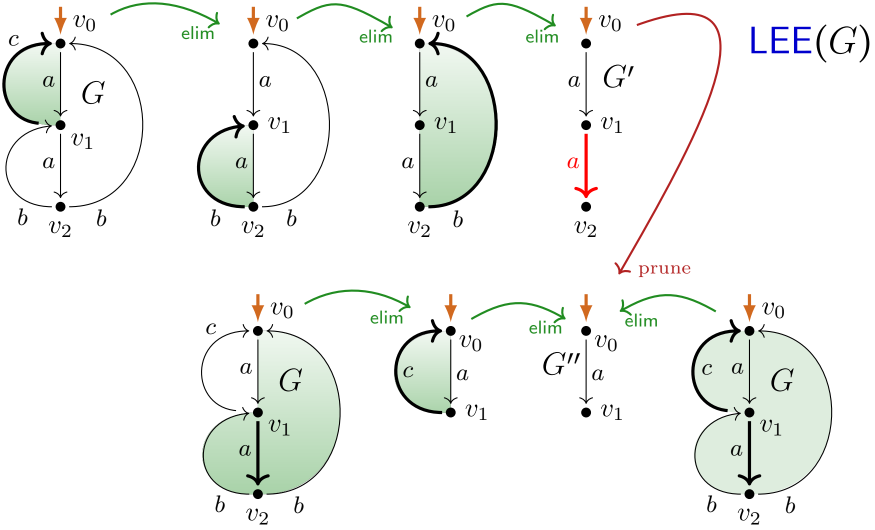


Joining an $\rightarrow_{\mathrm{elim}}$ fork with a pruning $\rightarrow_{\mathrm{prune}}$ step



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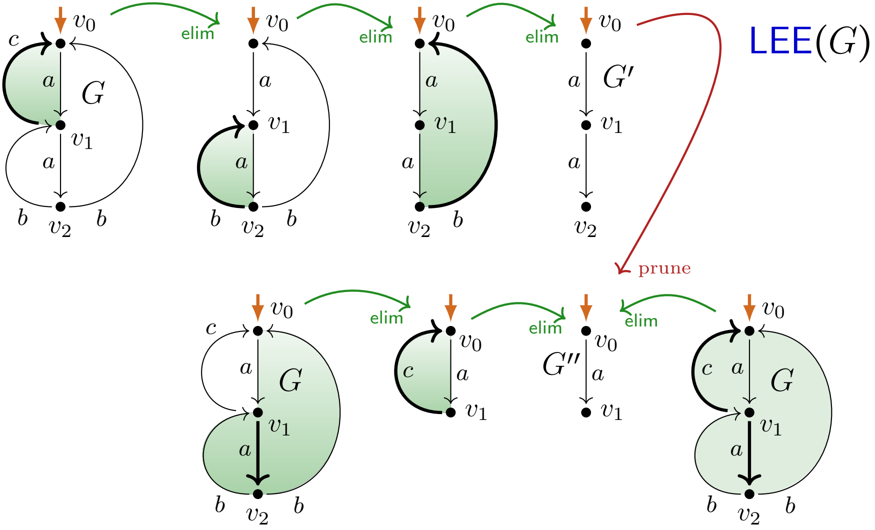

Overview

- Loop elimination $\rightarrow_{\text{elim}}$ in process graphs
  - Milner's process interpretation P of regular expressions
  - Loop existence and elimination property LEE
    LEE $\triangleq$ interpretations of $(*/\mathbb{1})$ reg. expr's
  - $\rightarrow_{\text{elim}}$ is not confluent
- Completion of loop elimination $\rightarrow_{\text{elim}}$ with pruning steps $\rightarrow_{\text{prune}}$
  - 'critical pair' lemma for 'bi-loop' elimination steps
  - use of decreasing diagrams
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‘Critical Pair Lemma’

Lemma (joining $\rightarrow_{\mathrm{elim}}$ forks)

Forks $G_2 \leftarrow_{\mathrm{elim}} G \rightarrow_{\mathrm{elim}} G_1$ in which loop subgraphs $G{\downarrow}_*^{v_2,T_2}$, $G{\downarrow}_*^{v_1,T_1}$ of G are eliminated, can be joined with $\rightarrow_{\mathrm{prune}}$ steps in the three situations:

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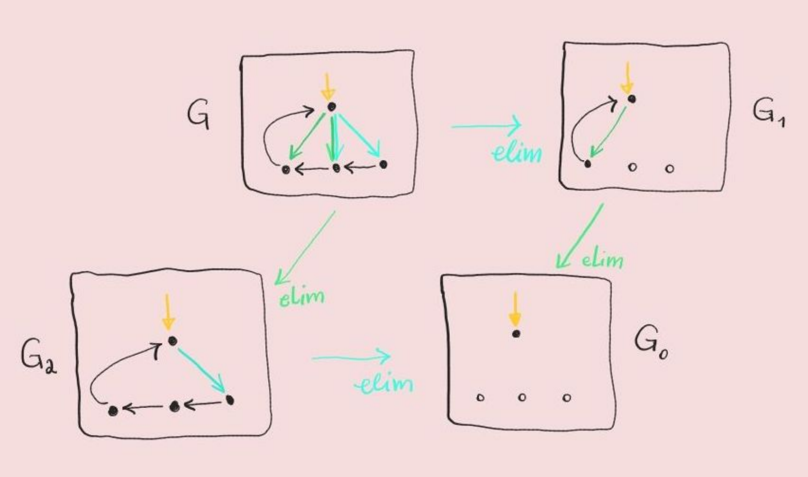
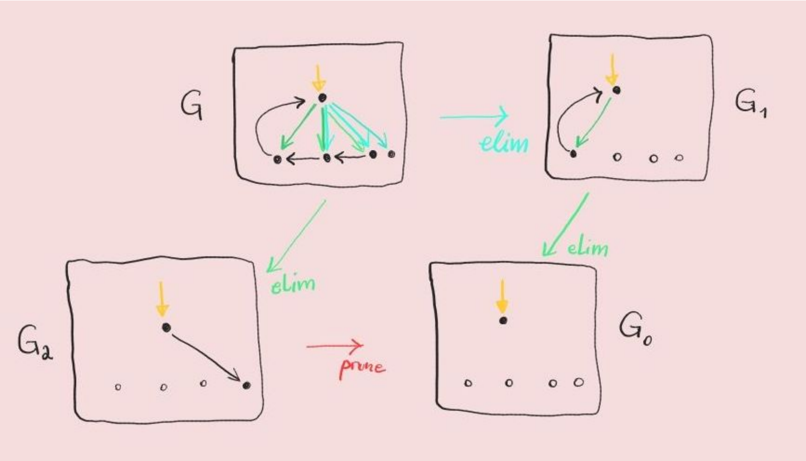
Root-overlapping loop subgraphs (example a)



Root-overlap of loop subgraphs (example b)



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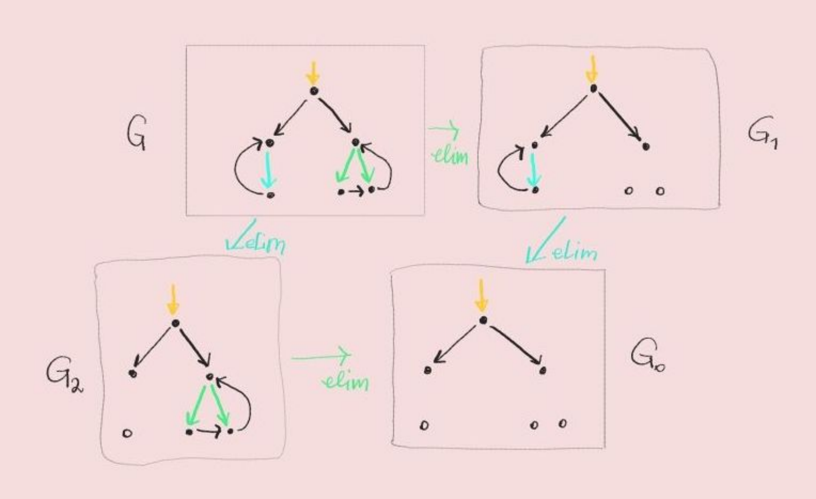
Parallel loop subgraphs (example)



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Lemma (joining $\to_{\text{elim}}$ forks)

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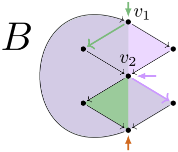

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bi-loop
w.r.t v_1, v_2
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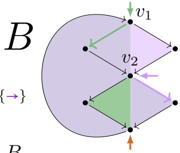
loop subgraphs of B



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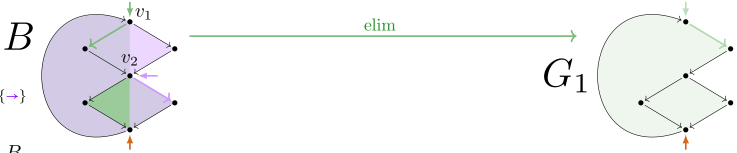

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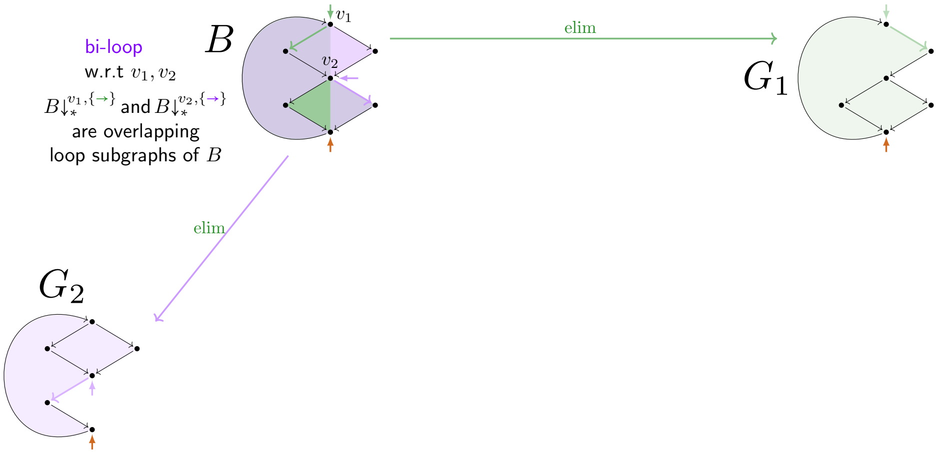

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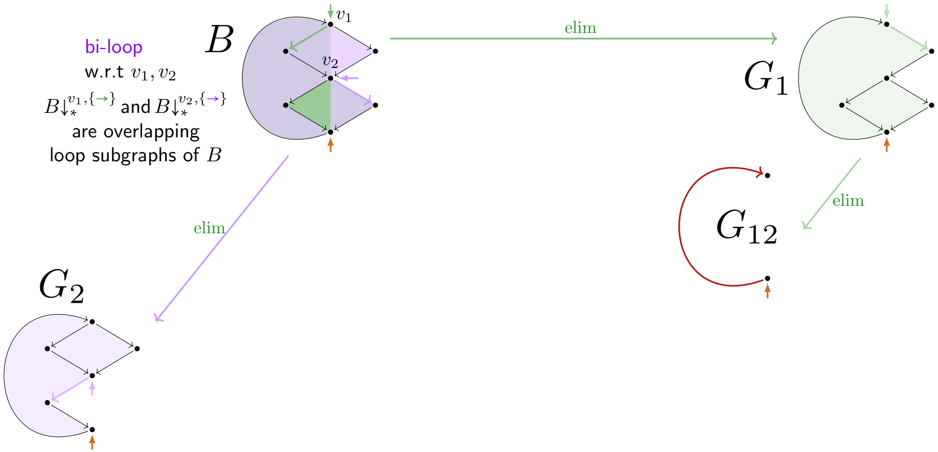

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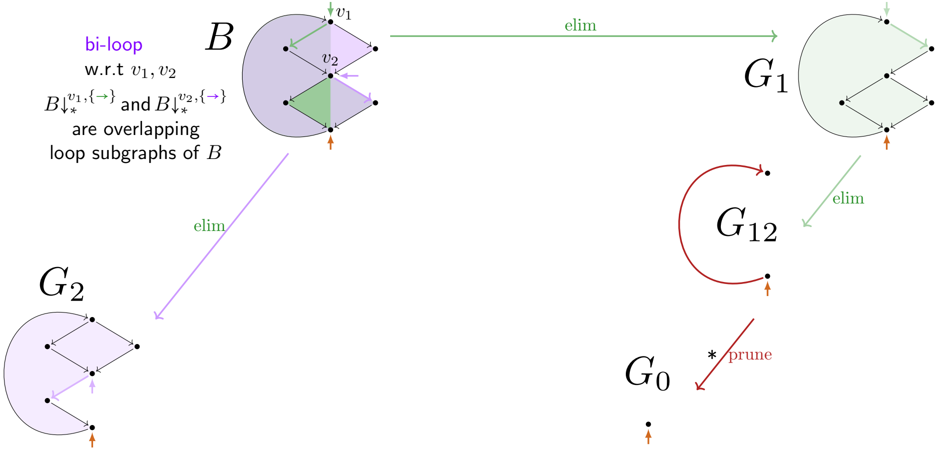

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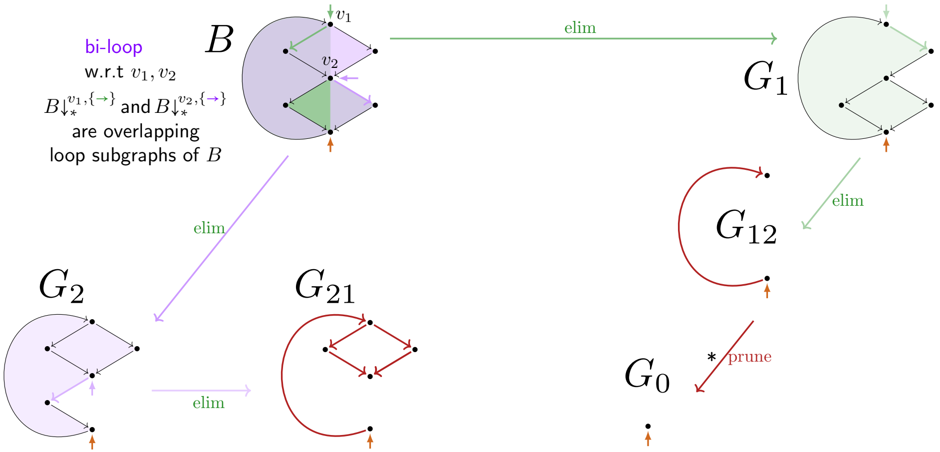

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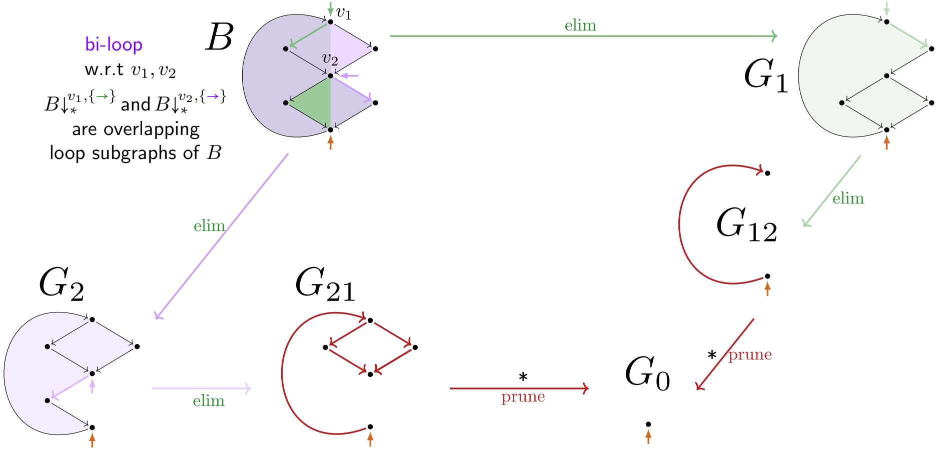

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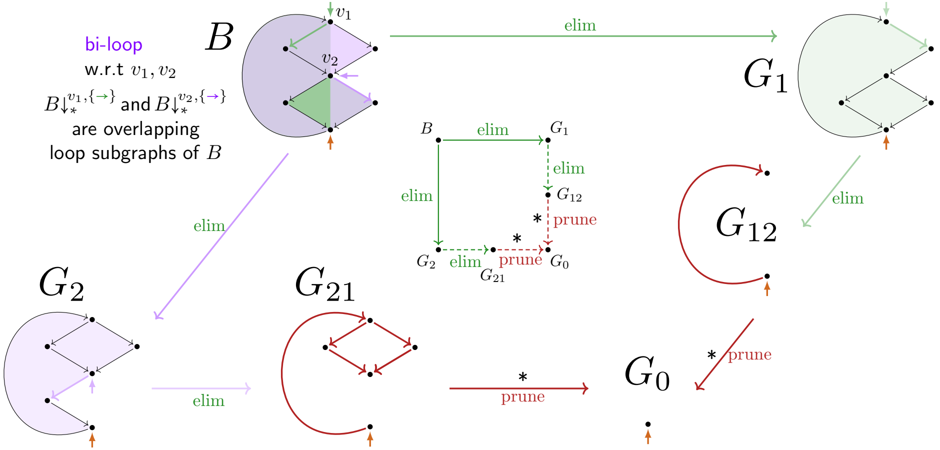

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Elementary diagrams

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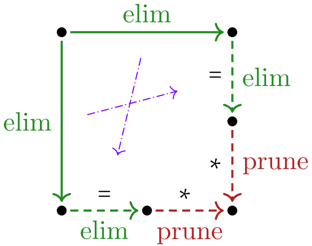

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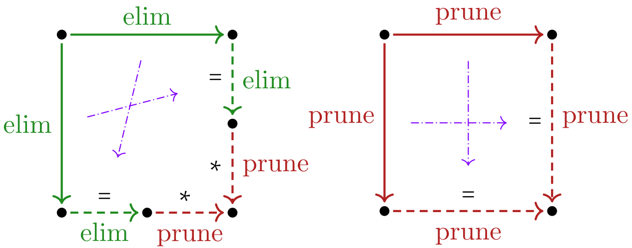

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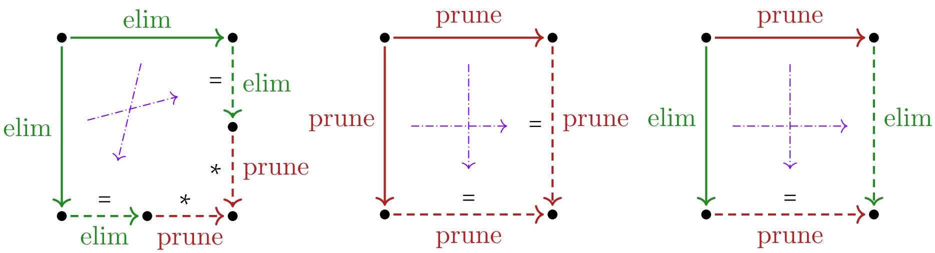

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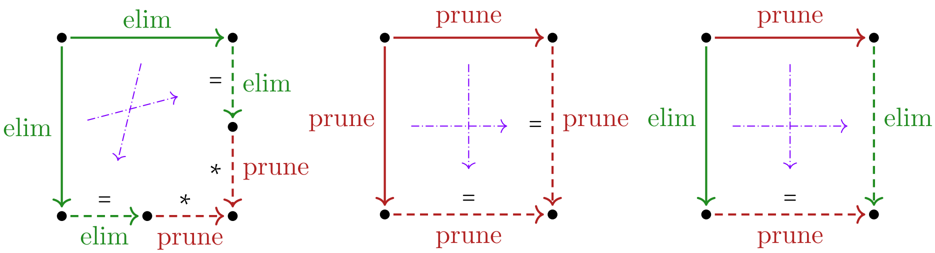

Theorem

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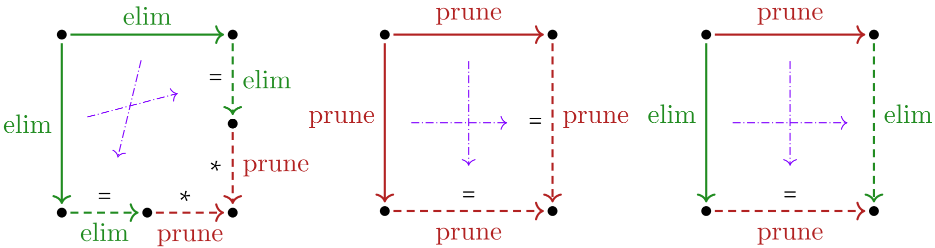

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Proof.

By the Decreasing Diagrams Theorem (van Oostrom~'94/de Bruijn). $\square$

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Efficient decidability of LEE

Corollary (of confluence of $\rightarrow_{\mathrm{ep}}$)

For every finite process graph G, TFAE:

(i) LEE(G): there is an $\rightarrow_{\mathrm{elim}}$ normal form of G <u>without</u> infinite trace.

(ii) The unique $\rightarrow_{\mathrm{ep}}$ normal form of G does <u>not</u> enable an infinite trace.

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Theorem

For finite graphs G, LEE is decidable in $\leq \mathrm{O}(|G|^3) \subseteq \mathrm{PTIME}$.

Proof

Due to (i) $\Rightarrow$ (iii) in the Corollary, to decide LEE(G) it suffices to construct any $\rightarrow_{\mathrm{elim}}$ rewrite sequence $G \rightarrow^*_{\mathrm{elim}} G_0 \not\rightarrow_{\mathrm{elim}}$ to normal form G_0, and check if $\neg\infty(G_0)$ holds. That is possible in $\leq \mathrm{O}(|G|^3)$ time. $\square$

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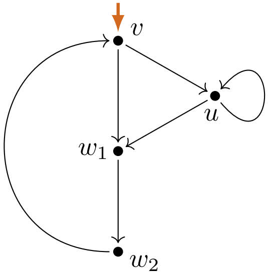

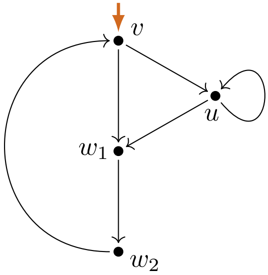

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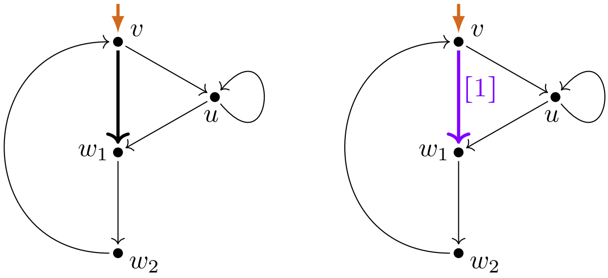

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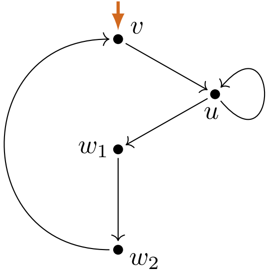

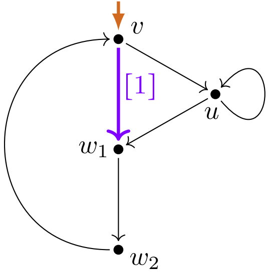

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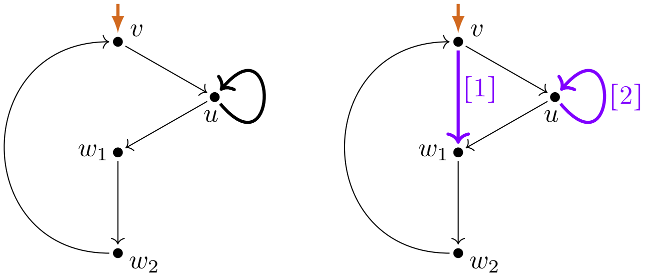

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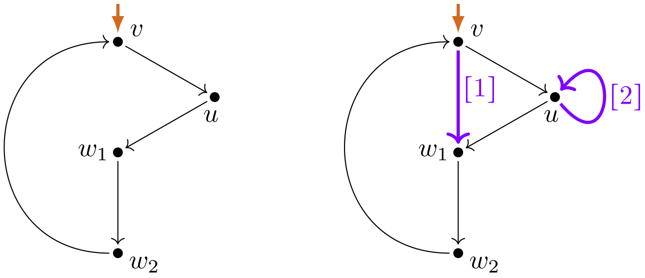

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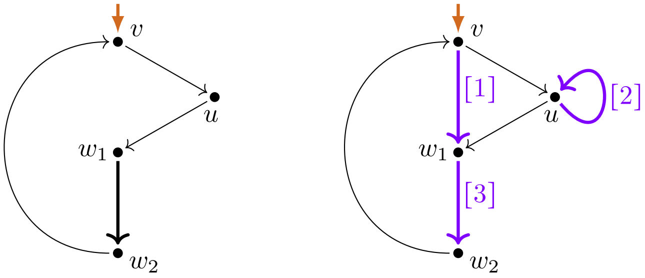

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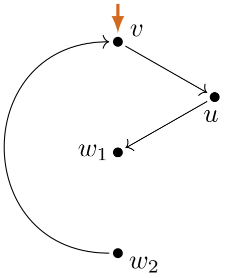

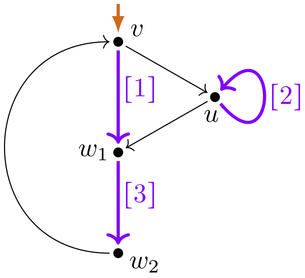

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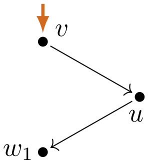

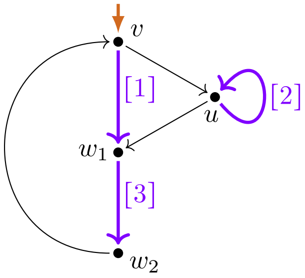

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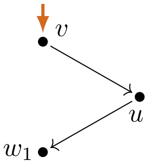

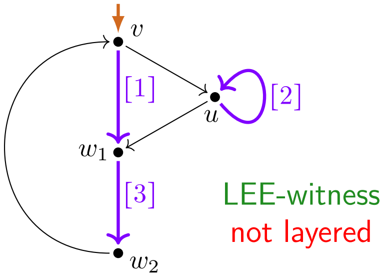

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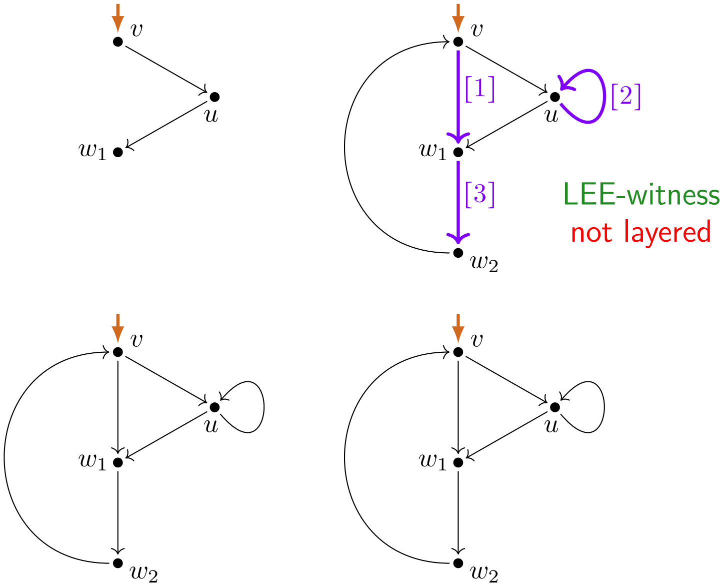

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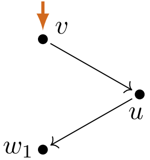

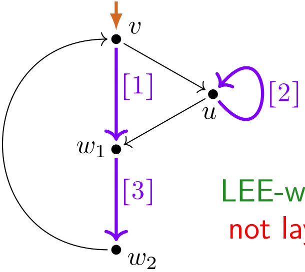

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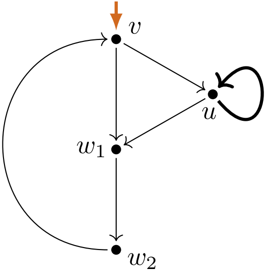

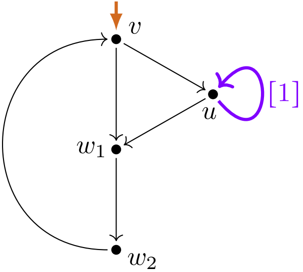

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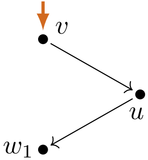

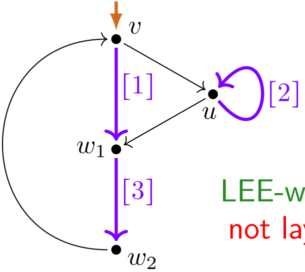

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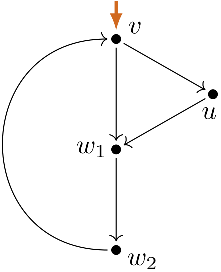

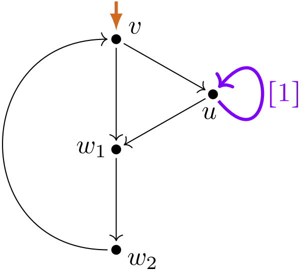

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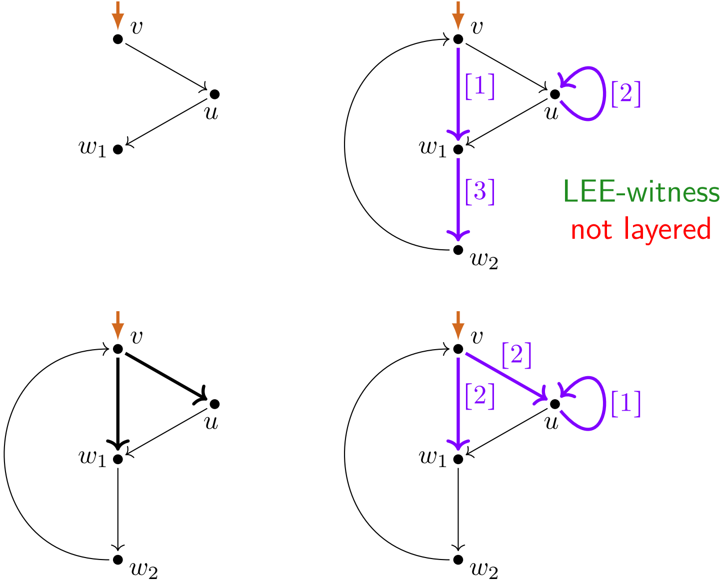

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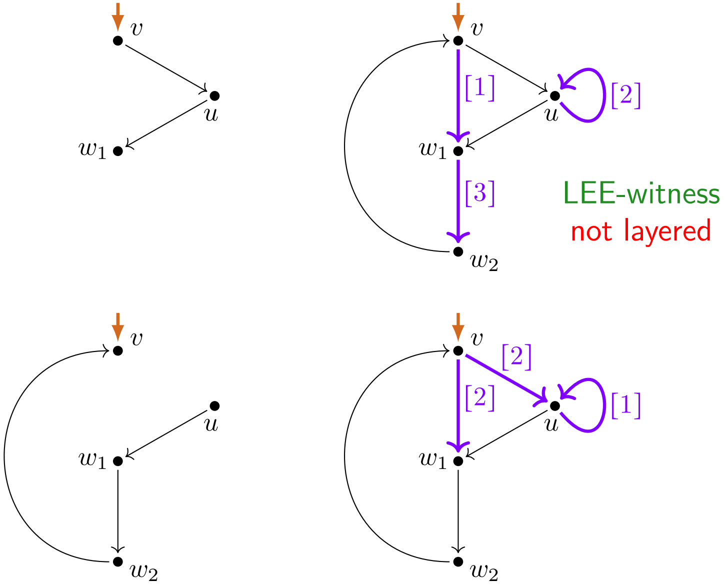

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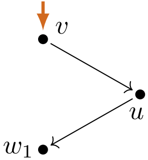

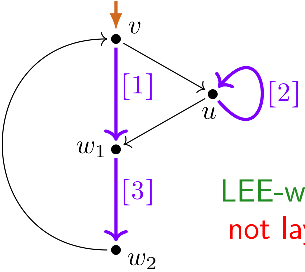

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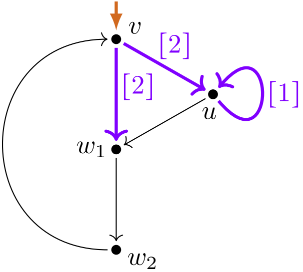

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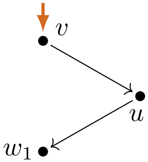

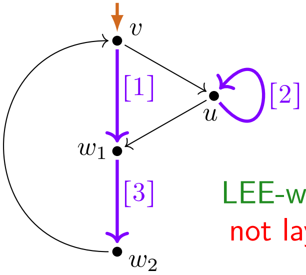

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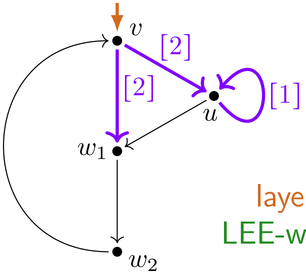

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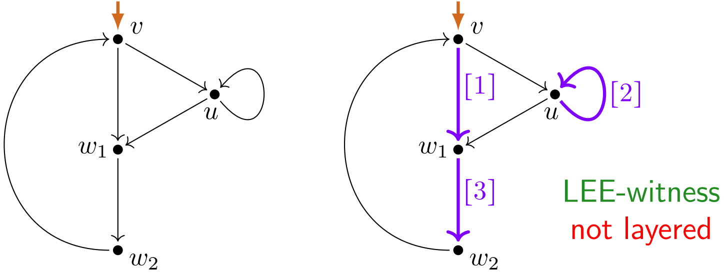

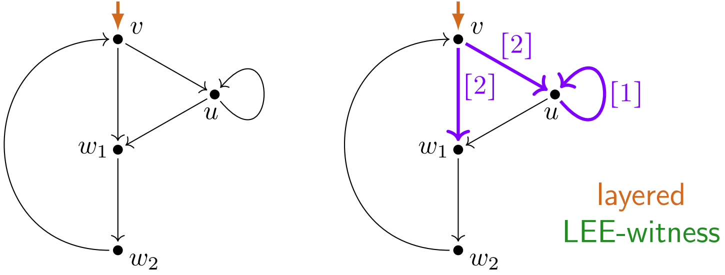

Layered Loop Elimination Condition LLEE

Layered loop elimination is defined on vertex-marked graphs $^{\circ}G$ in order to:

- mark vertices in the loop bodies of eliminated loops,
- permit the elimination of loop subcharts only at unmarked vertices.

Definition

The Layered Loop Existence & Elimination Condition LLEE for graphs G:

$$\mathsf{LLEE}(G) \;:\Longleftrightarrow\; \exists\, {}^{\bullet}G_0 \Big(\big({}^{\circ}G \overset{\circ}{\rightarrow}{}^{*}_{\mathrm{elim}} {}^{\bullet}G_0 \overset{\circ}{\not\rightarrow}_{\mathrm{elim}} \big) \wedge \neg\infty({}^{\bullet}G_0) \Big) .$$

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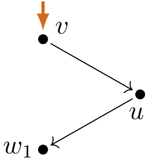
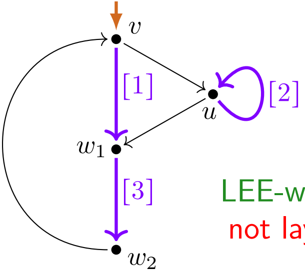
Theorem

$\mathsf{LLEE}(G) \Leftrightarrow \mathsf{LEE}(G)$ for all graphs G.

Making Layered





LEE-witness
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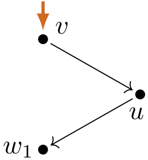

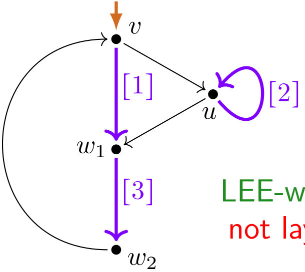

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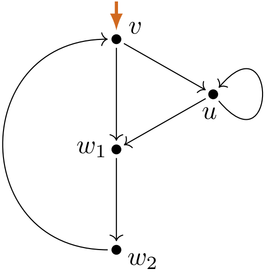

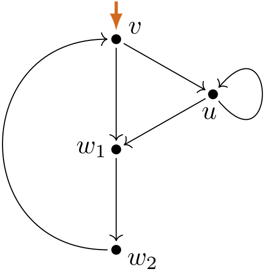

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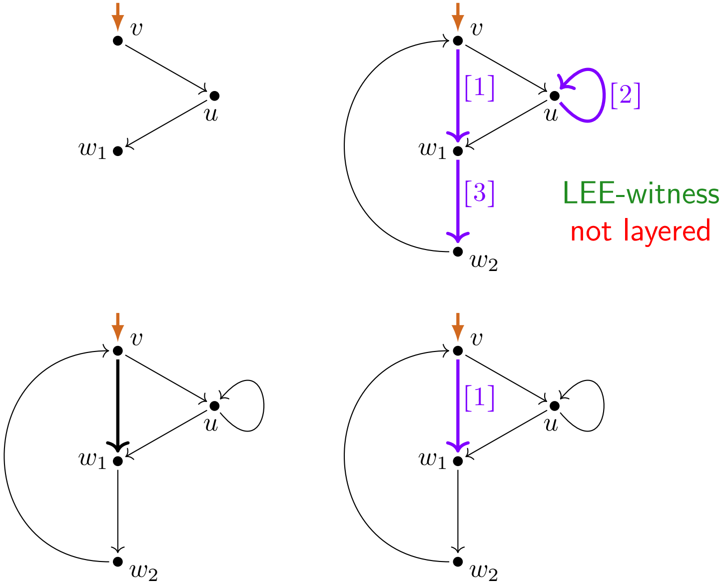

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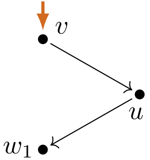

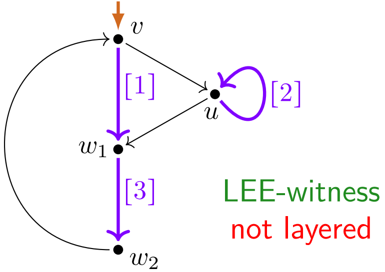

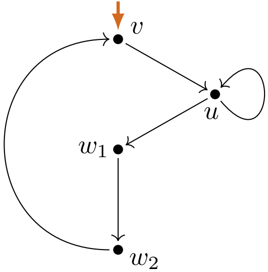

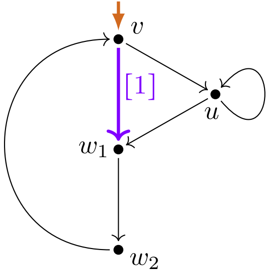

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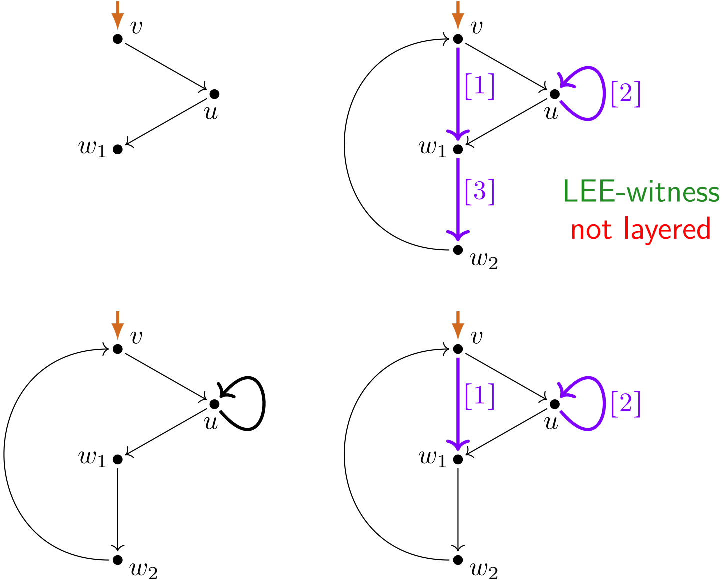

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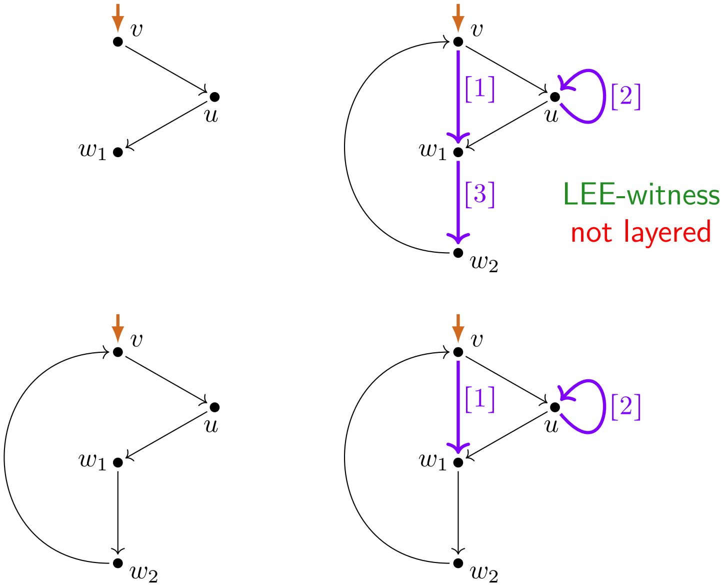

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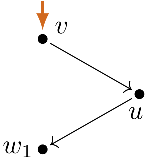

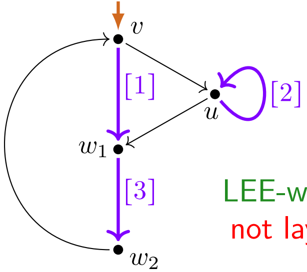

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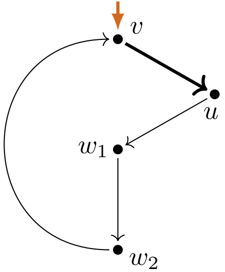

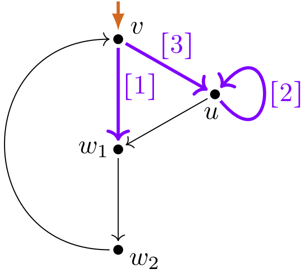

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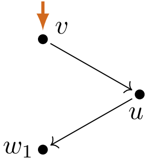

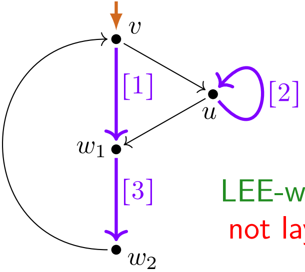

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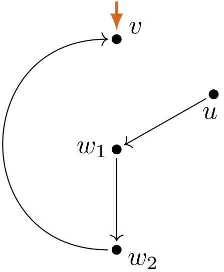

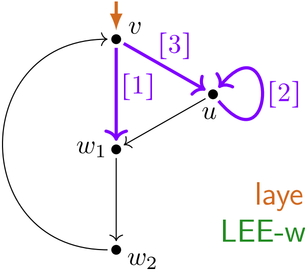

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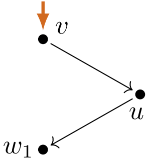

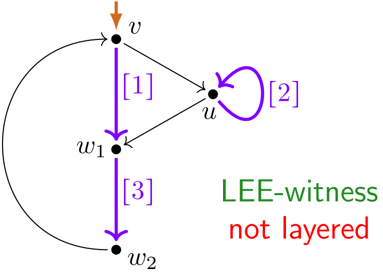

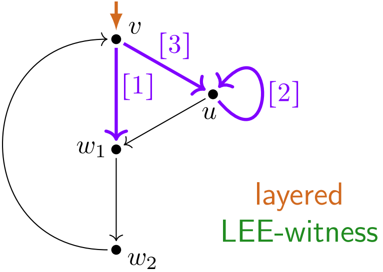

Further questions

(Q1) How efficient ($\leq O((n+k)^3)$) can LEE be decided?

- To try: characterisation of LEE as flow-graph co-reducibility (Laarakker–Kappé, TERMGRAPH 2026).

(Q2) Is 1-LEE decidable in polynomial time?
(1-LEE $\hat{=}$ generalised loop elimination, which is not confluent.)

(Q3) How can, for a graph G with LEE, any two regular expressions e_1, e_2 with $P(e_1) = P(e_2) \simeq G$ that are extracted from maximal $\rightarrow_{\mathrm{elim}}$ rewrite sequences from G be proved equivalent with respect to $[\![\cdot]\!]_P$?

(Q4) Can the Decreasing Diagrams Method be adapted to show pre-/post-ponement? (Here postponement of $\overset{\circ}{\rightarrow}_{\mathrm{elim}}$ over $\rightarrow_{\mathrm{elim}}$ is closely connected to confluence of $\rightarrow_{\mathrm{ep}}$.)

- Vincent van Oostrom referred me to his IWC 2020 article.

Summary

- Loop elimination $\rightarrow_{\mathrm{elim}}$ in process graphs
  - Milner's process interpretation P of regular expressions
  - Loop existence and elimination property LEE
    LEE = compact process interpretations of $(*/\cancel{1})$ reg. expr's
  - $\rightarrow_{\mathrm{elim}}$ is not confluent
- Completion of loop elimination $\rightarrow_{\mathrm{elim}}$ with pruning steps $\rightarrow_{\mathrm{prune}}$
  - 'critical pair' lemma for joining elimination forks $\leftarrow_{\mathrm{elim}} \cdot \rightarrow_{\mathrm{elim}}$
  - use of decreasing diagrams for $\rightarrow_{\mathrm{ep}} = \rightarrow_{\mathrm{elim}} \cup \rightarrow_{\mathrm{prune}}$
- Application 1: LEE is decidable in polynomial time
- Application 2: Layered LEE = LEE

IFIP-1.6 and TERMGRAPH talks

TERMGRAPH (last Sunday)

CG: *Towards a Characterisation of the Graph Structure of Process Interpretations of Regular Expressions*

Ext. Abstract `https://clegra.github.io/lf/pi-struc-TG-26.pdf`

Slides `https://clegra.github.io/lf/TG-26.pdf`

- LEE $\triangleq$ image of $P^{\bullet}|_{RExp^{(*/\cancel{1})}}$
- 1-LEE $\triangleq$ image of $P^{\bullet}$

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IFIP-1.6 meeting (last Saturday)

CG: *Finding Small Steps for Hard Problems, in Work on the Process Interpretation of Regular Expressions*

Slides https://clegra.github.io/lf/IFIP-1_6-2026.pdf

- Reducing for Graphs Representing Regular Expressions under Bisimilarity

Resources

- Slides/extended abstract on `clegra.github.io`
  - slides: `.../lf/IWC-26.pdf`
  - extended abstract: `.../lf/le-conf-IWC-26.pdf`
- CG: Milner's Proof System for
  Regular Expressions Modulo Bisimilarity is Complete
  - LICS 2022, arXiv:2209.12188, poster.
- CG: The Image of the Process Interpretation of Regular Expressions
  is Not Closed under Bisimulation Collapse
  - arXiv:2303.08553, 2021/2023.
- CG, Wan Fokkink: A Complete Proof System for
  1-Free Regular Expressions Modulo Bisimilarity,
  - LICS 2020, arXiv:2004.12740, video on youtube.
- CG: Structure-Constrained Process Graphs for
  the Process Semantics of Regular Expressions
  - TERMGRAPH 2020, EPTCS 334, arXiv:2012.10869.
- CG: Modeling Terms by Graphs with Structure Constraints
  - TERMGRAPH 2018, EPTCS 288, arXiv:1902.02010.